



MEGN 300: Spring 2026

Design Challenge Three: PID Easy Bake Oven

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Introduction

Our goal was to use a PID control system and rudimentary materials—cardboard, aluminum foil, and a 55W halogen bulb—to construct and operate a miniaturized oven capable of baking cookie dough at a user-defined temperature (350°F). The flow diagram pictured below in Figure 1 illustrates the manner the control system will process sensor data and output a corresponding controller response.

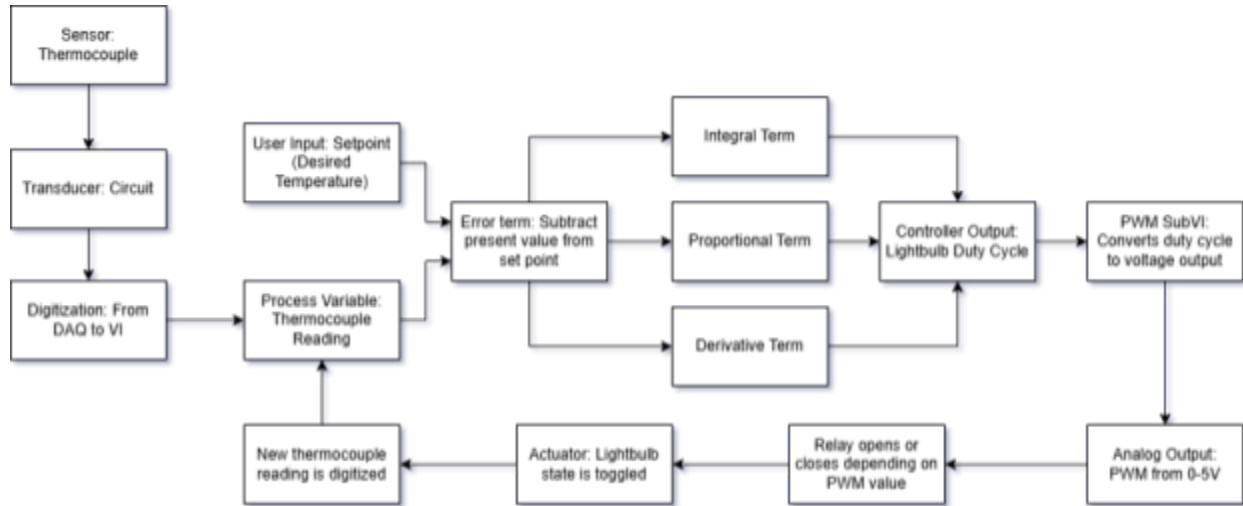


Figure 1: LabVIEW and PID flow diagram

First, the oven should measure its internal temperature with a thermocouple placed inside the heating chamber. This thermocouple will produce a voltage that the DAQ will receive by way of our circuit. It will be digitized by the DAQ and sent to a PID controller, which will use the user-defined temperature and PID gain variables to determine the duty cycle of the 55W bulb. This duty cycle will be converted to a PWM input, which will control the 55W bulb by way of a relay that can toggle it on and off. The oven will then take another thermocouple voltage reading, and the cycle will repeat until the user turns the oven off.

Design Calculations

To ensure that the oven could maintain a consistent internal cooking temperature, an approximate thermal resistance for each wall would need to be calculated. Given the oven's cubed form, all walls should have roughly the same thermal resistance. The area, thickness, and air gap for each wall is defined below.

$$A_{side} = h * w = 4in * 4in = 16in^2 \quad L_{air} = 0.5in \quad L_{cb} = 1.68mm$$

Using online data [1], the following values for the thermal conductivity of corrugated cardboard and an air gap were collected:

$$k_{cb} = 0.064 \frac{W}{m \cdot K}$$

$$k_{air} = 0.026 \frac{W}{m \cdot K}$$

These conductivity values were plugged into two equations for thermal resistance to calculate the thermal resistance of both the air gap and cardboard. The equations show that a larger air gap/wall thickness and a smaller thermal conductivity/area contributes to a greater thermal resistance.

$$R_{th,air} = \frac{L_{air}}{k_{air} \cdot A_{side}} = 47.32 \frac{K}{W}$$

$$R_{th,cb} = \frac{L_{cb}}{k_{cb} \cdot A_{side}} = 2.543 \frac{K}{W}$$

Thermal resistance can be modelled like electrical resistors in series. By summing them, a total resistance for one wall was calculated.

$$R_{th,wall} = 2 \cdot R_{th,cb} + R_{th,air} = 52.406 \frac{K}{W}$$

Only five of the six walls were taken into consideration for modelling purposes, as we assumed the wall making contact with the table would have a much greater effective thermal resistance than the other five. All five walls can be modelled like resistors in parallel, which will produce a total resistance equal to the resistance of one wall divided by five—a result of the equivalent thermal resistance for each wall. This yielded the following value for the total thermal resistance of the oven:

$$R_{th,total} = \frac{R_{th,wall}}{5} = 10.481 \frac{K}{W}$$

Lastly, the equation below was used to calculate the oven's heat loss at an internal temperature of 350°F.

$$\Delta T = 350 - 70 = 280 K$$

$$q_{loss} = \frac{\Delta T}{R_{th,total}} = 26.715 W$$

As shown above, our theoretical heat loss is less than half of the 55W of heat produced by the bulb. This results in the factor of safety calculated below.

$$FoS = \frac{55 W}{q_{loss}} = 2.05$$

Our oven design is therefore expected to maintain the desired internal temperature.

Methodology

Our Easy Bake oven was constructed by insulating a 4x4 cardboard box with aluminum foil and reflective tape. This box was placed inside of a larger 5x5 box in order to minimize heat loss and prevent damage to any components outside of the larger box. A halogen lightbulb was used as our actuator heating element and was inserted through both boxes and sealed with reflective tape. The halogen bulb duty cycle was managed via a Songle 5V relay. A thermocouple sensor was used to determine the internal temperature of the inner box. This sensor was inserted into the inner box via a small cutout through both layers. Once the cookie was inserted into the smaller box, both boxes were securely sealed off with painters and reflective tape ensuring minimal heat loss and improving safety. Table 1 lists all materials used in the design of our oven.

Bill of Materials	
4"x4" cardboard box	1x
5"x5" cardboard box	1x
Aluminum foil tape	~12 inches
Painter's tape	~18 inches
55W halogen bulb	1x
Songle SRD-05VDC-SL-C Relay	1x
Aluminum foil	~28 in length

Table 1: Bill of Materials

We selected a Songle SRD-05VDC-SL-C relay as our switching device for its reliable and relatively simple mechanical switching mechanism. According to the Tongling data sheet, the max allowable voltage is 110 VDC and the max power is 240W which is well above our use of 55W making it ideal for this project [2]. Figure 2 shows how the switching device is used with the DAQ and power supply units.

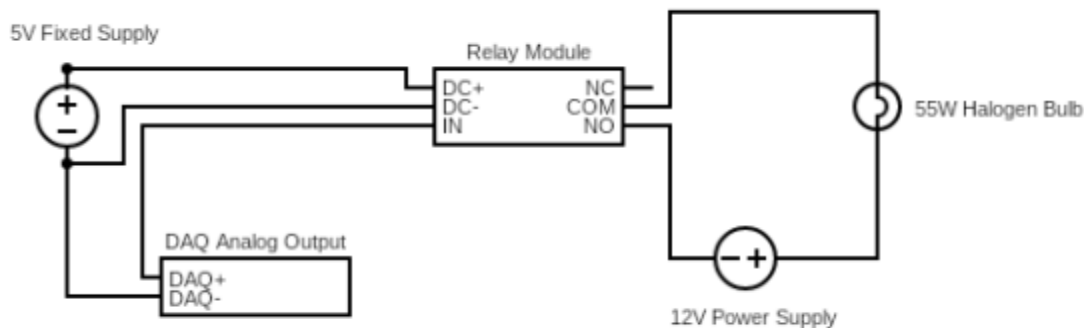


Figure 2: Circuit Diagram

Figure 3 describes the front panel of LabView VI allowing us to manage the K constants for the PID controller, control the setpoint when baking the cookie, and including a stop button to shut down the system when the cookie is finished baking .

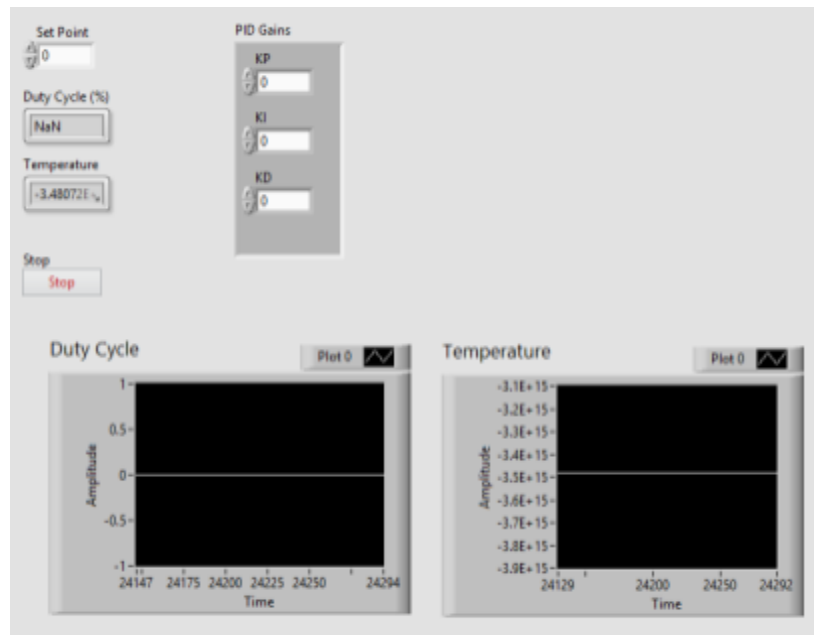


Figure 3: LabView VI Front Panel

The LabView block diagram is shown in figure 4 which provides an overview of our code logic. The block diagram includes the PID SubVI (detailed in Figure 1A), thermocouple sensor channel, and switching function to effectively control the duty cycle of the halogen bulb. PID gains of $K_p=10$, $K_d=0.25$, and $K_i=0.25$ were used, per prior tuning exercises.

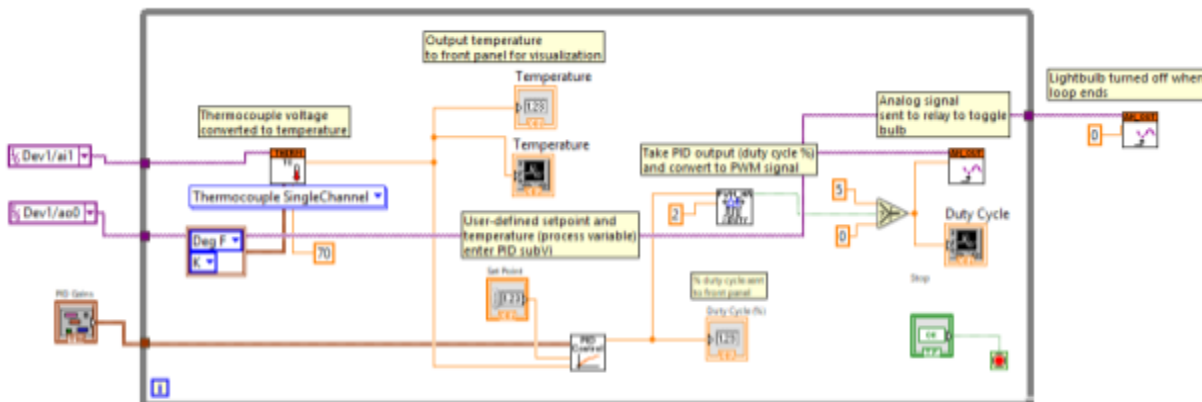


Figure 4: LabView VI Block Diagram

Results & Discussion

Using the dimensions and materials that were previously discussed, the oven was successfully constructed to specifications (shown below in Figure 5).



Figure 5: Finalized oven containing uncooked dough

After closing the oven, the control system was activated. A timer was set for 10 minutes, per the standard for most pre-made cookie doughs. Observing the VI's front panel showed a surprisingly quick climb up to the setpoint. As the temperature approached the setpoint, the PID system began to toggle the bulb to slow the rate of heating.

While there was a minor overshoot of approximately 5 degrees, the system eventually reached a steady-state temperature of just under 350°F. The duty cycle at this temperature appeared to hover around a value of 40%. After letting the timer run its course, the oven was carefully opened up to reveal a well-done cookie, shown below in Figure 6.



Figure 6: Cookie dough, post-cooking

While the cookie appears overcooked, most of the char was actually superficial damage. The underside and interior of the cookie was near perfect, as shown in Figure 7. As such, it can be assumed that the char was a result of the cookie's close proximity to the bulb, resulting in uneven cooking. An improved design might place the heating bulb further above or, like a true oven, far below the cookie. This would likely require a larger heating chamber, which may take more time to heat.



Figure 7: Cooked cookie, underside

Using our duty cycle value of 40% and bulb power (55W) we can estimate the actual heat loss that occurred during the baking process:

$$\text{Duty Cycle} = \frac{q_{\text{loss}}}{P_{\text{bulb}}}$$

$$q_{\text{loss}} = 0.4 * 55W = 22W = 22 \text{ J/s}$$

Based on our calculations, we found that our actual heat loss was approximately 22 J/s. This value is about 4 J/s less than our theoretical calculations. We accounted for the conductivity of the aluminum foil in our theoretical calculations but did not account for the heat reflection effect the aluminum foil has. The addition of the foil tape to seal the edges and cardboard cookie stand also likely contributed to a higher insulation factor. This resulted in a lower actual heat loss value for the oven. Overall, our oven performed well and heated up quicker than expected helping validate these findings.

Conclusion

Using the materials provided, we were able to successfully construct a miniaturized oven that could maintain a consistent internal temperature by way of proper insulation and a PID control system. Our estimated heat loss was rather close to what was actually observed during the cooking process (26 J/s vs. 22 J/s), showing that our calculations could accurately (though conservatively) predict the real life performance of the oven. Though slightly overcooked, our cookie was brought to the intended temperature of 350°F, which was maintained by the PID control system for the duration of cooking. The oven could be improved by optimizing the distance between the heating element and the material being cooked, resulting in a uniform distribution of heat through the subject.

Works Cited

[1] Engineers Edge, "Thermal Properties of Non-Metals," *Engineers Edge*, 2026. [Online]. Available: https://www.engineersedge.com/heat_transfer/thermal_properties_of_nonmetals_13967.htm. [Accessed: April 29, 2026].

[2] Ningbo Songle Relay Co., Ltd., "SRD Series Relay Datasheet," SRD-05VDC-SL-C, n.d. [Online]. Available: [SRD-05VDC-SL-C pdf](#), [SRD-05VDC-SL-C Description](#), [SRD-05VDC-SL-C Datasheet](#), [SRD-05VDC-SL-C view :: ALLDATASHEET ::](#) [Accessed: April 29, 2026].

Appendix

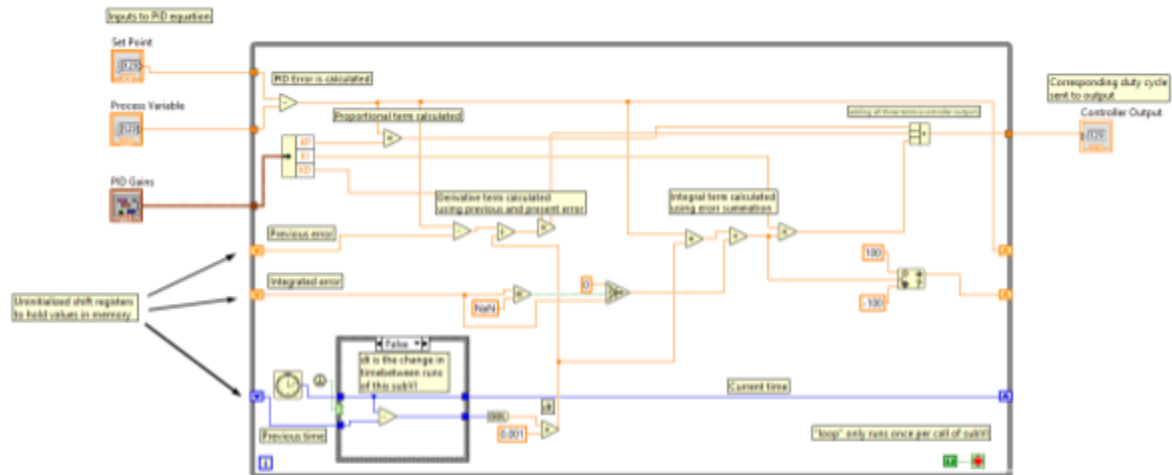


Figure 1A: PID SubVI